AI Assisted Coding

Assignment 9.5

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Task_01 :

Consider the following Python function:
def reverse_string(text):
return text[::-1] Task:
1. Write documentation in:
o (a) Docstring o (b) Inline comments o (c) Google-style documentation
2. Compare the three documentation styles.
3. Recommend the most suitable style for a utility-based string library.

Prompt:
""" generate Docstring inline comments and google style documentation for following code """

Outputs:

Inline comment& Docstring documentation

```
09 > prblm1.py > reverse_string
"""  generate Docstring inline comments and google style documentation for following code  """
def reverse_string(s): # function to reverse a string
    """Reverse a given string.

    Args:
        s (str): The string to be reversed.

    Returns:
        str: The reversed string.
    """
    return s[::-1] # return the reversed string using slicing
```

BLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

PyDoc Documentation

```
TERMINAL                                                                    python

PS C:\Users\Vivek\OneDrive\Desktop\3-2\AI-Assisted-Coding> python -m pydoc Lab_09.prblm1

NAME
    Lab_09.prblm1 - generate Docstring inline comments and google style documentation for following code

FUNCTIONS
    reverse_string(s)
        Reverse a given string.

        Args:
            s (str): The string to be reversed.

        Returns:
            str: The reversed string.
-- More  --
```

```
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc quest1
    ~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
KeyboardInterrupt
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc -w  quest1
wrote quest1.html
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc -p 3000
Server ready at http://localhost:3000/
Server commands: [b]rowser, [q]uit
```

Ln 12, Col 1 Spaces: 4 UTF-8 CRLF

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Python 3.13.7 [tags/v3.13.7:bcee1c3, MSC v.1944 64 bit (AMD64)]
Windows-11

quest1

generate Docstring inline comments and google style documentation for following code

Functions

reverse_string(s)

```
Reverse a given string.

Args:
    s (str): The string to be reversed.

Returns:
    str: The reversed string.
```

Task_02

Problem 2: Password Strength Checker Consider the function: def check_strength(password):

return len(password) >= 8

Task:

1. Document the function using docstring, inline comments, and Google style.
2. Compare documentation styles for security-related code.
3. Recommend the most appropriate style.

Output:

Inline and Docstring Documentation:

```
""" generate Docstring inline comments and google style documentation for following code """
def check_strength(password):# function to check the strength of a password
    """Check if the given password is strong.

    A strong password is defined as having at least 8 characters.

    Args:
        password (str): The password to be checked."""

    return len(password) >= 8 # return True if the password is strong, otherwise return False
# Example usage:
password = "my_secure_password"
if check_strength(password):
    print("The password is strong.")
else:
    print("The password is weak.")
```

pyDoc Documentation:

```
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc quest2
The password is strong.
Help on module quest2:

NAME
    quest2 - generate Docstring inline comments and google style documentation for following code

FUNCTIONS
    check_strength(password)
        Check if the given password is strong.

        A strong password is defined as having at least 8 characters.
```

0 2 Ln 16, Col 35 Spaces: 4

```
    ~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
KeyboardInterrupt
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc -w  quest2
The password is strong.
wrote quest2.html
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc -p 3000
Server ready at http://localhost:3000/
Server commands: [b]rowser, [q]uit
server>
```

0 2 Ln

Pydoc: Index of Modules | Pydoc: Index of Modules | Pydoc: Error - get?key=prblm1 | Pydoc: module quest1 | Pydoc: module quest2

localhost:3000/quest2.html

Python 3.13.7 [tags/v3.13.7:bcee1c3, MSC v.1944 64 bit (AMD64)]
Windows-11

Module Index : Topics : Keywords
Get

quest2

c:\users\vivek\onedrive\desktop\lab_09\qu

generate Docstring inline comments and google style documentation for following code

Functions

check_strength(password)
Check if the given password is strong.

A strong password is defined as having at least 8 characters.

Args:
    password (str): The password to be checked.

Data

password = 'my_secure_password'

Task-03:

Problem 3: Math Utilities Module Task:
1. Create a module math_utils.py with functions:
o square(n) o
cube(n) o
factorial(n)
2. Generate docstrings automatically using AI tools.
3. Export documentation as an HTML file.

Code:

(Inline and docstring documentation)

(square and cube)

quest3.py > ...

```python
""" generate Docstring inline comments and google style documentation for Crea
o square(n)
o cube(n)
o factorial(n) """
def square(n):
    """Calculate the square of a number.

    Args:
        n (int or float): The number to be squared.

    Returns:
        int or float: The square of the input number.
    """
    return n ** 2 # return the square of the number
def cube(n):
    """Calculate the cube of a number.

    Args:
        n (int or float): The number to be cubed.

    Returns:
        int or float: The cube of the input number.
    """
    return n ** 3 # return the cube of the number
```

```python
    return n ** 3 # return the cube of the number
def factorial(n):
    """Calculate the factorial of a number.

    Args:
        n (int): The number to calculate the factorial for. Must be a non-negative integer.

    Returns:
        int: The factorial of the input number.

    Raises:
        ValueError: If n is a negative integer.
    """
    if n < 0:
        raise ValueError("Factorial is not defined for negative integers.")
    elif n == 0 or n == 1:
        return 1 # return 1 for factorial of 0 and 1
    else:
        result = 1
        for i in range(2, n + 1):
            result *= i # multiply result by each integer from 2 to n
        return result # return the factorial of the number
```

PyDoc

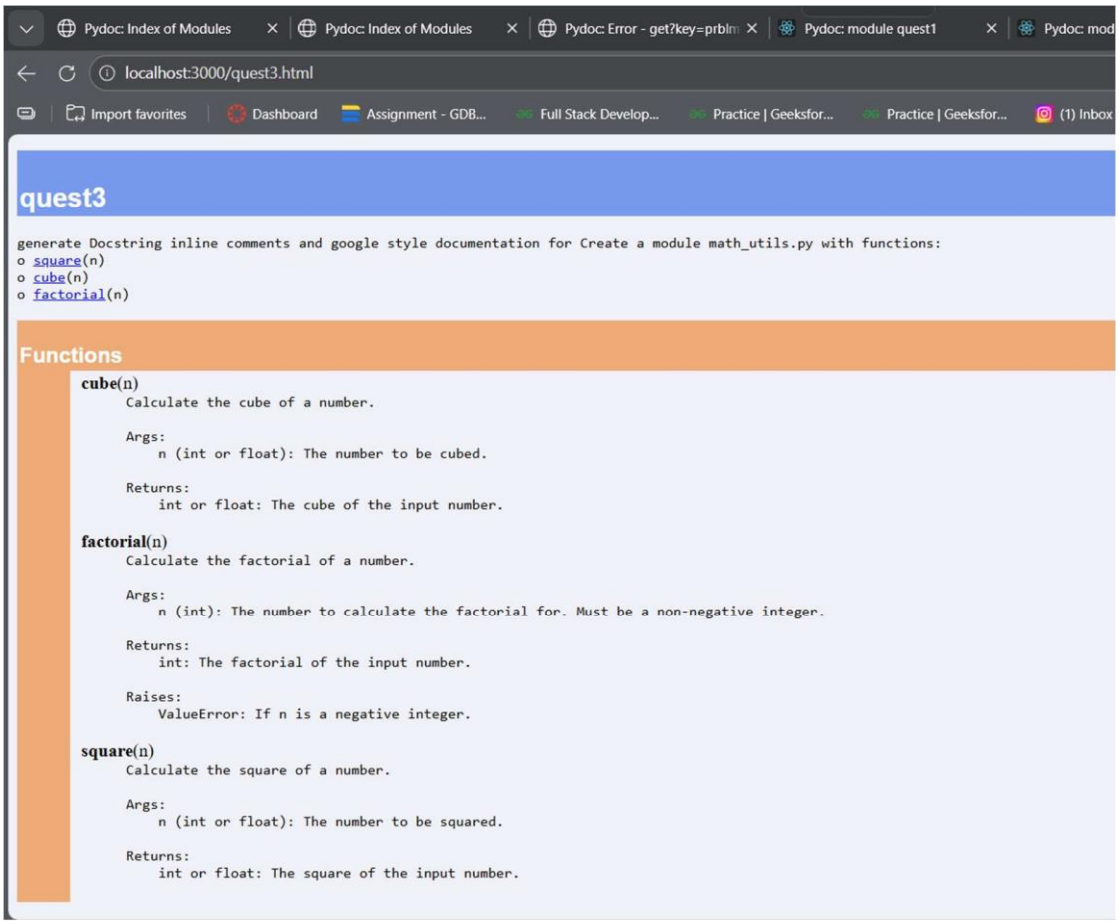
```
q
q …
DESCRIPTION
    generate Docstring inline comments and google style documentation for Create a module math_utils.py with functions:
    o square(n)
    o cube(n)
    o factorial(n)

FUNCTIONS
    cube(n)
        Calculate the cube of a number.

        Args:
            n (int or float): The number to be cubed.
-- More --
```

0 Ln 46, Col 5 Spaces: 4 UTF-8 CRLF Python Select Int

```
        ~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
KeyboardInterrupt
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc -w  quest3
wrote quest3.htm
PS C:\Users\Vive                           ab_09> python -m pydoc -p 3000
Server ready at http://localhost:3000/
Server commands: [b]rowser, [q]uit
server>
```

Follow link (ctrl + click)

0 ⚠ 0



Task_4:

Problem 4: Attendance Management Module Task:

1. Create a module attendance.py with functions:

o mark_present(student)

o mark_absent(student) o

get_attendance(student)

2. Add proper docstrings.

3. Generate and view documentation in terminal and browse

Code:

Inline and DocString Documentation:

```python
attendance = {} # dictionary to store attendance records
def mark_present(student):
    """Mark a student as present.

    Args:
        student (str): The name of the student to mark as present.
    """
    attendance[student] = 'Present' # mark the student as present in the attendance dictionary
def mark_absent(student):
    """Mark a student as absent.

    Args:
        student (str): The name of the student to mark as absent.
    """
    attendance[student] = 'Absent' # mark the student as absent in the attendance dictionary
def get_attendance(student):
    """Get the attendance status of a student.

    Args:
        student (str): The name of the student to check attendance for.

    Returns:
        str: The attendance status of the student ('Present', 'Absent', or 'Not Recorded').
    """
    return attendance.get(student, 'Not Recorded') # return the attendance status of the student, or
```

PyDoc Documentation:

```
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc quest4
Present
Absent
Not Recorded
Help on module quest4:

NAME
    quest4

DESCRIPTION
    generate Docstring inline comments and google style documentation for Create a modu
tions:
    Task:
    1. Create a module attendance.py with functions:
     mark_present(student)
     mark_absent(student)
     get_attendance(student)
     Add proper docstrings.

FUNCTIONS
    get_attendance(student)
^CTraceback (most recent call last):
```

```
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc -w  quest4
Present
Absent
Not Recorded
wrote quest4.html
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc -p 3000
Server ready at http://localhost:3000/
Server commands: [b]rowser, [q]uit
```

quest4

```
generate Docstring inline comments and google style documentation for Create a module math_utils.py with functions:
Task:
1. Create a module attendance.py with functions:
 mark_present(student)
 mark_absent(student)
 get_attendance(student)
 Add proper docstrings.
```

Functions

get_attendance(student)

```
    Get the attendance status of a student.

    Args:
        student (str): The name of the student to check attendance for.

    Returns:
        str: The attendance status of the student ('Present', 'Absent', or 'Not Recorded').
```

mark_absent(student)

```
    Mark a student as absent.

    Args:
        student (str): The name of the student to mark as absent.
```

mark_present(student)

```
    Mark a student as present.

    Args:
        student (str): The name of the student to mark as present.
```

Data

attendance = {'Alice': 'Present', 'Bob': 'Absent'}

Task_05:
Problem 5: File Handling Function
Consider the function: def
read_file(filename):
with open(filename, 'r') as f:
return f.read() Task:
1. Write documentation using all three formats.
2. Identify which style best explains exception handling.
3. Justify your recommendation

Inline and DocString Documentation

```
""" Consider the function:
def read_file(filename):
with open(filename, 'r') as f:
return f.read()
Task:
1. Write documentation using Docstring inline comments and google style documentation for the above function.
"""
def read_file(filename):
    """Read the contents of a file.

    Args:
        filename (str): The name of the file to be read.

    Returns:
        str: The contents of the file as a string.

    Raises:
        FileNotFoundError: If the specified file does not exist.
        IOError: If there is an error reading the file.
    """
    with open(filename, 'r') as f: # open the file in read mode
        return f.read() # return the contents of the file as a string
# Example usage:
try:
    content = read_file('example.txt')
    print(content)
except FileNotFoundError:
    print("The specified file was not found.")
except IOError:
    print("An error occurred while reading the file.")
```

PyDoc:

```
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc quest5
Vivek Lakum

Help on module quest5:

NAME
    quest5

DESCRIPTION
    Consider the function:
-- More  --
```

```
KeyboardInterrupt
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc -w  quest5
Vivek Lakum

wrote quest5.html
PS C:\Users\Vivek\OneDrive\Desktop\lab_09> python -m pydoc -p 3000
Server ready at http://localhost:3000/
Server commands: [b]rowser, [q]uit
server>
```

Python 3.13.7 [tags/v3.13.7:bcee1c3, MSC v.1944 64 bit (AMD64)]
Windows-11

quest5

```
Consider the function:
def read_file(filename):
with open(filename, 'r') as f:
return f.read()
Task:
1. Write documentation using Docstring inline comments and google style documentation for the above function.
```

Functions

read_file(filename)

```
    Read the contents of a file.

    Args:
        filename (str): The name of the file to be read.

    Returns:
        str: The contents of the file as a string.

    Raises:
        FileNotFoundError: If the specified file does not exist.
        IOError: If there is an error reading the file.
```

Data

content = 'Vivek Lakum \n'